The Paleography of Pharmacy: Tironian Notae and the Pharmaceutical Shorthand Hypothesis for the Voynich Manuscript

The investigation into the Voynich Manuscript (MS 408), carbon-dated to the period between 1404 and 1438, has increasingly shifted from the search for a simple substitution cipher toward the analysis of specialized procedural notation systems. The hypothesis that the manuscript represents a unique, high-density shorthand developed by an apothecary or a medical practitioner finds compelling support in the historical practices of medieval pharmaceutical Latin. In this context, the text is viewed not as a hidden language, but as a professional jargon encoded through a system of abbreviations and sigla that draws deeply from the ancient tradition of Tironian Notes. Computational analyses yielding a z-score of 9.4 on 500 random trials suggest a statistical regularity in the manuscript's character distribution that mirrors the efficiency of established shorthand systems rather than the stochastic variability of natural language or the patterned redundancy of traditional ciphers.[1] This report examines the mechanics of medieval pharmaceutical abbreviations, the survival of Tironian shorthand in the 15th century, and the paleographical evidence connecting these traditions to the structural anomalies of MS 408.

Medieval Pharmaceutical Latin and the Professionalization of Shorthand

The development of a specialized pharmaceutical language was a direct result of the professionalization of medicine in Europe, centered largely around the School of Salerno. The 11th and 12th centuries saw the emergence of standardized pharmacopoeias, most notably the *Antidotarium Nicolai*, which established a rigid framework for the preparation of compound medicines.[4] For the medieval apothecary, Latin was the essential medium of communication, but it was a Latin stripped of rhetorical flourish and reduced to its functional essence through a dense layer of abbreviations.[6]

The Mechanics of Apothecary Abbreviations

Medieval scribes and apothecaries utilized a system of "notae" or sigla that allowed them to compress complex recipes into a few lines of parchment.[8] This was not merely an aesthetic choice but an economic necessity, as the high cost of vellum dictated a maximalist approach to information density.[8] The system was built on three primary pillars: truncation, contraction, and the use of special signs. Truncation involved writing only the initial letters of a word, while contraction omitted intermediate letters, often leaving the first and last characters.[7]

In the pharmaceutical context, this meant that the most common instructions—the verbs and measurements—were reduced to single-letter or two-letter sigla. The *Antidotarium Nicolai*, for instance, standardized the Salernitan system of weights, which utilized a specific set of symbols to ensure precision across different regions and languages.[4] The use of these symbols created a professional boundary; the recipe became a "code" that could only be deciphered by those trained in the trade, thereby maintaining quality control and professional secrecy.[6]

Standard Instructions and Preparation Verbs

The structure of a medieval recipe followed a predictable procedural sequence, usually beginning with the imperative command to "take" specific ingredients.[14] The following table outlines the standard abbreviations for the verbs and instructions that formed the backbone of the apothecary's notation.

Abbreviation	Latin Term	English Instruction	Function in Recipe
Rx / ℞ / Rp.	*Recipe*	Take	The starting point of every formula.[14]
M. / m.	*Misce*	Mix	Direction to combine ingredients.[18]
coq.	*Coque*	Boil	Thermal processing instruction.[18]
ter. / t.	*Tere*	Rub / Grind	Mechanical processing in a mortar.[20]
col.	*Cola*	Strain	Removal of solids from a liquid.[21]
ft.	*Fiat*	Let it be made	Declaration of the final product form.[18]
sig.	*Signa*	Write / Label	Directions for administration.[16]

aa / ana	*Ana*	Of each	Indicates equal parts of ingredients.[16]
q.s.	*Quantum sufficit*	A sufficient quantity	Used for solvents or binders.[16]
m.d.s.	*Misce, detur, signetur*	Mix, give, sign	Standard closing formula.[18]

The usage of *ana* (often written as *aa* with a bar above) is particularly significant in the context of MS 408's repetitive patterns. In a pharmaceutical list, the repetition of *ana* serves to link disparate ingredients to a single quantity, a structural feature that may explain some of the "word" repetitions observed in Voynichese.[16]

The Tironian System and the 15th Century Shorthand Tradition

The Tironian Notes, the first standardized Roman shorthand system, underwent a massive expansion during the medieval period.[25] While often associated with the Carolingian Renaissance of the 8th and 9th centuries, the system never truly disappeared; instead, it became "fossilized" within the professional scripts of the law and medicine.[7] By the 15th century, the time of the Voynich Manuscript's creation, the Tironian system had expanded to over 13,000 signs, as cataloged in the *Commentarii Notarum Tironianarum*.[25]

Survival in the Veneto and Lombardy Regions

Evidence suggests that Northern Italy, specifically the Veneto and Lombardy regions, remained a vibrant center for the use of shorthand and idiosyncratic ligatures well into the 15th century.[3] Humanistic scribes in cities like Padua and Venice were experimenting with formal scripts that incorporated complex abbreviations to save space.[30] The "7" for *et* and the "9" prefix for *con-* are the most recognizable remnants, but in professional pharmaceutical circles, the adoption of Tironian-derived signs was likely more extensive.[26]

A key aspect of Tironian Notes is their composite nature. Signs were often ligatures where a basic root mark was modified by auxiliary strokes to represent different tenses, cases, or related concepts.[25] This mirrors the structure of Voynich "words," which often consist of a stable "core" accompanied by a predictable set of prefixes and suffixes.[24] The computational finding of a high statistical connection to Tironian systems (z-score 9.4) suggests that the MS 408 scribe was utilizing a similar economy of symbols where each glyph might represent an entire syllable or common pharmaceutical term.[1]

The Role of the Commentarii Notarum Tironianarum

Wilhelm Schmitz's 1893 edition of the *Commentarii* reveals the depth of this system. The "sixth commentarius" specifically deals with concept-based signs, including some that are clearly relevant to the scientific and medical world.[34] While a single "dictionary" of pharmaceutical Tironian notes has not survived, the existence of these specialized lists indicates that different professions—lawyers, clerics, and presumably physicians—developed their own subsets of shorthand.[27]

The 15th-century Mantuan ciphers further demonstrate this hybridity. In documents from 1401 and 1450, we find the Tironian "9" shape and the "Rx" symbol used alongside arbitrary shapes that stand in for whole syllable groups like *ab, ac, ad, af, ag*.[33] This demonstrates that the concept of "nomenclators"—codebooks where specific symbols represent whole words or common syllables—was well-established in the very region and time frame associated with the Voynich Manuscript.[33]

Comparative Paleography of MS 408: The Gallows Characters

One of the most defining features of the Voynich Manuscript is the set of "gallows" characters: *p, f, k,* and *t* in the EVA transcription system.[1] These tall, ornate characters often appear as line-initial or paragraph-initial markers, and their behavior suggests they are more than simple letters.[1]

Gallows as Procedural Markers

In medieval paleography, the use of ornate, oversized characters at the beginning of sections was a standard way to signal a change in subject or a new step in a process.[36] These characters often functioned as "capitula" or section markers.[36] In a pharmaceutical context, the gallows characters might denote specific stages of a recipe:

- **EVA-p and EVA-f:** These double-loop and single-loop "P-shaped" characters strongly resemble the *Rx* (recipe) symbol or the *per* (through/by) abbreviation.[14]
- **EVA-k and EVA-t:** These "pi-shaped" characters bear a functional resemblance to notarial signs used in Lombardy to mark the start of a formal investiture or instruction.[33]

The "coverage" property observed in MS 408—where a gallows character's crossbar extends over subsequent letters—is reminiscent of the way medieval scribes used a single abbreviation mark (a macron or tilde) to indicate that all the characters beneath it were part of an abbreviated cluster.[1] If the gallows are viewed as functional shorthand for commands like *Coque* (Boil) or *Misce* (Mix), their placement at the start of paragraphs makes perfect sense as a procedural heading for a specific pharmaceutical preparation.[5]

Statistical Profile and Numerical Hypothesis

Some researchers have proposed that the gallows characters might represent numbers, specifically denoters for hundreds or thousands, which would be essential for documenting the quantities of ingredients in a large-scale pharmacopoeia.[40] This theory is consistent with the way apothecaries had to carefully measure substances like *Mithridate*, which could contain upwards of 50 different ingredients.[41]

Voynich Grapheme (EVA)	**Paleographical Parallel**	**Potential Shorthand Meaning**
p / f	*Recipe / Per*	"Take" or a specific measurement unit.[14]
k / t	*Notarial Sigla*	Section marker or procedural command (*Coque*).[36]
9-shape	*Tironian 'con-'*	Common prefix in pharmaceutical terms (*confectio*).[32]
7-shape	*Tironian 'et'*	Conjunction linking ingredients in a list.[25]
□-shape	*Abbreviation for '-us'*	Terminal marker for ingredient names (*aloe-us*).[32]

Abbreviation Patterns for Ingredients and Prepositions

The "how" of pharmaceutical abbreviation was governed by the need to identify ingredients quickly. Common substances were rarely written in full; instead, a standardized set of abbreviations was used, many of which persisted from the Salernitan era into the early modern period.[19]

Ingredient Notation and Common Sigla

In a manuscript like the *Antidotarium Nicolai*, the ingredients are the primary focus.[4] The following list represents the actual abbreviation conventions used by apothecaries for common materials.[41]

- **Aqua (aq.):** Water, the universal solvent. In prescriptions, it often appeared with a modifier, such as *aq. ros.* (rose water) or *aq. font.* (spring water).[22]
- **Oleum (ol.):** Oil, used in ointments (*unguenta*) and electuaries. Often abbreviated as *ol. rosat.* (rose oil).[41]
- **Mel (mel.):** Honey, the primary excipient for electuaries in the "sugar-honey pharmacy" of Salerno.[5]
- **Sal (sal.):** Salt, often specified as *sal nitri* or *sal commune*.[47]
- **Cera (cer.):** Wax, used for *cerates* and stiff ointments.[48]
- **Aloe (al.):** A common cathartic, often yielding Socotrine or Cape varieties.[42]
- **Cinnamomum (cinam.):** Cinnamon, a frequent aromatic additive in compound medicines.[41]
- **Mastic (mast.):** A resin used as a binder and for its therapeutic properties.[46]

The role of "material status markers" was also critical. An apothecary needed to know if an ingredient was *crudo* (raw), *pulverizatum* (powdered), or *distillata* (distilled). These were often indicated by superscript letters (e.g., ^{r} for *recipe*, ^{p} for *pulverizatum*) or by trailing shorthand marks like the Tironian symbols for *-um* or *-is*.[32]

Prepositional Agglutination and Structural Linguistic Shifts

A significant linguistic phenomenon in late medieval Latin—particularly in technical and medical texts—is the agglutination of prepositions and articles.[52] As Latin transitioned toward the vernacular, the use of case endings was increasingly replaced by prepositions like *in, cum, per, de,* and *ex*.[52] In shorthand, these prepositions were frequently fused to the following word, creating long "concatenated" strings.[53]

- **Cum (with):** Often represented by a backward "c" (□) fused directly to the noun.[32]
- **Per (by/through):** Indicated by a crossed "p" (□) that became a prefix.[32]
- **In (in):** Often denoted by a simple macron over the first letter of the following word.[8]

This agglutination may account for the "word" length in Voynichese. A single "word" in the manuscript might not be a lexical unit in the traditional sense, but a cluster representing a prepositional phrase, such as "with-water" (*cum-aqua*) or "by-boiling" (*per-coquere*).[24] The lack of traditional word boundaries in some shorthand systems further complicates the analysis, as the graphic space may not respect linguistic boundaries.[54]

Manuscript Analysis: From Salerno to the Veneto

To understand the shorthand of MS 408, it is essential to compare it to known 14th- and 15th-century Italian pharmaceutical manuscripts.

The Salernitan Heritage: Antidotarium Nicolai and Circa Instans

The *Antidotarium Nicolai* (12th century) and the *Circa Instans* (12th century) provided the foundational recipes.[4] In their earliest forms, they used standard Salernitan abbreviations.[4] However, by the 14th century, copies like BNF Latin 6823 (produced in Southern Italy) show an evolution toward more cursive hands and more dense abbreviation patterns.[49] This manuscript includes an unillustrated copy of the *Antidotarium Nicolai*, where recipes like *Athanasia* are written in a script that begins to prioritize speed over clarity.[49]

Balneological and Herbal Traditions in Northern Italy

The *De Balneis Puteolanis*, a famous set of balneological poems, is frequently cited as a parallel for the "balneological" section of the Voynich Manuscript.[37] These texts were often illustrated and utilized a specific set of technical terms related to thermal baths and their medicinal properties.[37]

In the 15th-century Veneto, the *Tractatus de Herbis* and the *Herbarium Carrarense* represent the pinnacle of this tradition.[29] These manuscripts were often "interleaved" over generations, with new labels and recipes added in cursive, highly abbreviated scripts.[29] The use of humanistic formal hands in Padua during the 1450s—such as those of Bartolomeo Sanvito—shows a move toward a "lapidary" serif style, but the everyday professional script of the apothecary (*mercantesca*) remained a dense, loop-filled shorthand designed for rapid notation.[30]

Shorthand in the "Small-Plants" and "Pharma" Sections

The "pharma" or small-plants section of MS 408 is particularly suggestive of a procedural manual.[55] In typical 15th-century herbals, these sections listed "simples" (single ingredients) and their "virtues" (therapeutic effects).[50] The labels in these sections often lack the "qo-qo" prefix found in the main text, which may suggest that labels represent simple nouns (ingredient names), while the running text represents the procedural instructions (imperative verbs and prepositional phrases).[24]

Modern Lexicographical Resources and Data Analysis

The search for the "actual" abbreviation conventions used by apothecaries relies on a handful of essential modern resources.

The Cappelli Dictionary and Pharmaceutical Latin

Adriano Cappelli's *Dizionario di Abbreviature* (1899) is the primary reference for medieval Latin paleography.[9] While it is a general dictionary, it covers a significant number of pharmaceutical and medical abbreviations found in Italian manuscripts of the 14th and 15th centuries.[7] Cappelli includes specific tables of "sigla" and "signs of abbreviation" that were common in Lombard and

Venetian documents.[33]

Databases and Digital Paleography

Modern databases for searching abbreviation patterns include the *Abbreviationes* web resource and the Unicode-based MUFI (Medieval Unicode Font Initiative) datasets.[9] These tools allow researchers to match Voynich glyphs against thousands of attested medieval abbreviations.[25]

The statistical connection between MS 408 and Tironian shorthand, as evidenced by z-score analysis, points to a system where characters have a high degree of predictability based on their neighbors—a hallmark of shorthand systems like Tironian Notes or modern stenography.[1] In such a system, the frequency of certain "bigrams" (two-character sequences) is extremely high because they represent common Latin syllables or pharmaceutical prefixes.[2]

Synthesis: The Apothecary Shorthand Hypothesis

The hypothesis that the Voynich Manuscript scribe was an apothecary who adapted Tironian shorthand for pharmaceutical notation provides a robust framework for understanding the manuscript’s unique features. The evidence from 15th-century Northern Italian medical manuscripts suggests that the profession relied on a high-density, secretive system of notation that was procedural in nature.

Procedural Efficiency vs. Simple Encryption

MS 408 does not behave like a simple substitution cipher; its linguistic structure is too regular, and its "vocabulary" is too repetitive.[24] However, it behaves precisely like a professional shorthand. In a pharmaceutical manual, the same commands (*Take, Mix, Boil*) and the same ingredients (*Water, Honey, Oil*) would appear thousands of times, leading to exactly the kind of low-entropy statistical profile observed in the manuscript.[2]

The gallows characters, with their ornate structure and positional regularity, are functional markers that mirror the notarial and pharmaceutical section signs of the Veneto and Lombardy regions.[33] The "benched" characters and repetitive prefixes represent the agglutination of prepositions and common prefixes (*con, per, pro*) that was standard in the technical Latin of the period.[32]

Conclusions for Decryption

The decryption of the Voynich Manuscript likely requires the reconstruction of a 15th-century pharmaceutical "nomenclator" based on the Tironian Notes tradition.[3] By mapping the manuscript’s glyphs to the established sigla for preparation verbs (*Recipe, Misce, Coque, Tere*), ingredients (*Aqua, Mel, Oleum, Sal*), and prepositions (*in, cum, per*), the "words" of MS 408 may reveal themselves not as a lost language, but as a highly efficient, professional shorthand for the compounding of medicine.

The connection to Tironian Notes, as cataloged by Schmitz and preserved in the professional scripts described by Cappelli, provides the historical "key" to this system.[12] The manuscript is a testament to the meticulous evolution of medical practice, ensuring that healing remained a structured, accountable, and professional process, even when recorded in a script that would remain an "elegant enigma" for centuries.[56]

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